

Yu-Ting Liao (Harry)

computational artist & designer — the body as an interface

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ABOUT

Computational artist and designer working on the **body as an interface**, across embodied sound, wearable and tabletop instruments, and real-time voice AI. Completed the MA Computational Arts at Goldsmiths in 2026 with **Solo Choir**, a real-time instrument that turns one singer into a choir in their own voice. Trained in fashion pattern cutting at UAL and in generative-AI media at Shih Hsin University, with five years of professional visual design experience in Taipei. Works across hardware prototyping, machine learning and creative code.

EDUCATION

Sep 2025 — Sep 2026

MA in Computational Arts — Goldsmiths, University of London

Thesis project: **Solo Choir** (see Research & Projects). Viva Aug 2026; final award expected Oct 2026.

Physical computing, embodied interaction, real-time audio and machine learning for creative practice (Raspberry Pi Pico, Arduino, Python, p5.js, Tone.js, TouchDesigner).

Sep 2022 — Jul 2024

MA in Digital Multimedia Arts — Shih Hsin University, Taipei

Graduate thesis: *Technology Inside AI Game Development and Application: An Example of Generative AI NPC*. Developed an AI NPC dialogue system in Unreal Engine 5.2 using GPT-4, integrating speech recognition, synthesis and sentiment analysis for context-aware, real-time NPC interaction.

Sep 2016 — Jun 2019

BA in Fashion Pattern Cutting — University of the Arts London

RESEARCH & SELECTED PROJECTS

2026

Solo Choir — MA thesis · real-time vocal instrument · Goldsmiths

Turns one singer into a choir: sing a phrase and the system answers with harmony voices re-synthesised in the singer's **own voice**, using real-time neural voice conversion on CPU.

Tabletop responsive instrument (four speakers, head model, boom microphone) shown with two wearable prototypes: an open-ear bone-conduction headband and an auxetic neck ring for feeling harmony in the body.

demo: vimeo.com/1223687812 · code: github.com/yvwHY/solo-choir · [case study](#)

2026

Fabri-Form — robotics · pattern cutting · Goldsmiths

A robotic pattern-cutting interface: a multimodal LLM reads a hand-drawn sketch, generates 2D cutting paths, and a Raspberry Pi Pico-driven laser traces them onto fabric.

2025

The Bonnet — wearable · computer vision · Goldsmiths

An anti-surveillance headpiece: a custom-trained YOLOv8 model detects a CCTV camera and an Arduino servo drops a fabric veil over the face.

2025

Gesture-Float — hardware · wireless control · Goldsmiths

A wireless flying unit steered by the arm: an arm-mounted Pico W streams hand pitch and roll over Wi-Fi (UDP) to a helium balloon unit, on hand-soldered circuits.

EXHIBITIONS

Aug 2026	Scream, Unsynced — MA Computational Arts Festival 2026, Goldsmiths, London Postgraduate degree show, St James Hatcham Building, 27–29 Aug 2026. Exhibited Solo Choir.
May 2026	Bye Default — Computational Arts Pop-Up, PSH Atrium, Goldsmiths, London Exhibited <i>Harmonizing!</i> , a browser-based live vocal-harmony instrument; designed the exhibition graphics and caption cards for all 37 works.

EXPERIENCE

Sep 2023 – Jun 2025	Visual Designer — Nice Crush Design House, Taipei Created visual assets, including digital advertisements, social media graphics, packaging and promotional materials. Developed marketing strategies to ensure cohesive visual communication across both online and offline platforms.
Jan 2021 – Sep 2022	Visual Designer — Digibionic, Taipei Designed and created visual assets for brand marketing. Managed marketing materials, promotions and product logistics. Supported marketing and operational tasks.
Dec 2020 – Jun 2021	Graphic Designer — AY-AY International, Taipei Created graphic assets and layouts for a wide mix of multimedia projects. Produced and maintained style guides.

SKILLS

physical computing	Raspberry Pi Pico W, Arduino, servo/motor control, circuit soldering, sensor fusion, bone-conduction and surface-exciter audio
audio / machine learning	real-time neural voice conversion, DDSP, pitch detection, Web Audio API, YOLOv8, OpenCV, Python, custom dataset curation (Roboflow)
creative code	p5.js, Tone.js, ml5.js, TouchDesigner, JavaScript
3d / game engine	Unreal Engine 5, Unity, 3D computer graphics
design	Adobe Illustrator, Photoshop, InDesign, After Effects; pattern cutting and garment construction

references and full documentation available on request · yvwhy.github.io · updated Sep 2026